

System developed to translate ancient 'dead' languages



In developing a system to help decipher lost languages, MIT researchers studied the language of Ugaritic, which is related to Hebrew and has previously been analysed and deciphered by linguists (Credit: S.R.K. Branavan)

A system developed at MIT's Computer Science and Artificial Intelligence Laboratory (CSAIL) is able to automatically decipher a lost language, without needing advanced knowledge of its relation to other languages.

The system relies on several principles grounded in insights from historical linguistics, such as the fact that languages generally only evolve in certain predictable ways. For instance, while a given language rarely adds or deletes an entire sound, certain sound substitutions are likely to occur.

The system's algorithm learns to embed language sounds into a multidimensional space where differences in pronunciation are reflected in the distance between corresponding vectors. This design enables them to capture pertinent patterns of language change and express them as computational constraints. The resulting model can segment words in an ancient language and map them to counterparts in a related language.

Hawk-inspired robot with movable wings is an agile long-distance flyer

A robot with wings has been developed that can fly more stably and nimbly than other flying robots—and it uses less power, extending flight time. The goal was to develop a drone that could fly long distances across cities, but manoeuvre around buildings and objects that it is likely to encounter, as multicopter drones can't fly long distances.

With wings and tail spread fully, the robot gains height. When it reaches top speed, the feather-like plates can be tucked in to become more aerodynamic, just like a bird.

The tail also moves up and down and side to side, enabling the robot to change its altitude quickly. Each wing can also retract or expand independently, as needed, to increase or reduce drag. By tucking in its wings and tail when travelling at its optimal speed of 9.6 metres per second, the drone uses 55.4 per cent less power than would be required to travel at that speed with its wings and tail fully open.



This hawk-like drone is an agile and efficient flyer (Credit: 2020 EPFL/Alain Herzog)

Cellulose could play an important role in future green electronics

Researchers from Uppsala University, Sweden and Ulsan National Institute of Science and Technology, Korea have explored the feasibility of cellulose in energy storage devices composed of conducting composites and thin layers of electro-active materials, as well as cellulose-based current collectors and functional separators.

Cellulose has been found to be an excellent material in the fabrication of electrodes as well as battery separators through a straightforward approach that deposits a thin layer of a conductive material on the surface of a cellulose sheet, thus converting the insulating paper into a flexible, conductive current collector. Further, cellulose is an attractive substrate for printed energy storage devices due to its superior mechanical

flexibility, thermal stability, and recyclability.

Cellulose derived from wood conducts differently compared to cellulose from algae or cotton, which requires further investigations for proper selection of the types of cellulose to be employed. A too low or a too high porosity would decrease the battery performance. The water content of the cellulose is another important parameter.

Thus, research on cellulose-based supercapacitors and batteries is very promising. Considering the properties of cellulose and the possibilities to employ existing large-scale manufacturing methods, cellulose-based energy storage systems will enable novel devices that will serve as complements to conventional technology.